Problem Set 1

Problem 1.

Suppose $X_1, \ldots, X_n$ are independent and identically distributed random variables from a Unif$(a, a+\theta)$ distribution, where the parameter a and $\theta > 0$ are unknown.

(a) **Maximum Likelihood Estimation:** Derive the Maximum Likelihood Estimator (MLE) for the parameter θ, denoted by $\hat{\theta}$.

(b) **Sampling Distribution:** Find the probability density function of $\hat{\theta}$.

(c) **Bias Analysis:** Determine whether $\hat{\theta}$ is an unbiased estimator of θ. If it is biased, calculate the bias.

(d) **Jackknife Bias Estimate:** Compute the jackknife estimate of bias for $\hat{\theta}$.

(e) **Bias Correction:** Obtain the bias-corrected estimator using the jackknife method.

(f) **Comparative Analysis:** Compare the bias of the original MLE with that of the jackknife-corrected estimator. Comment on why the jackknife is particularly effective for this "boundary-style" parameter.

Problem 2.

Let $X_1, \ldots, X_n$ be an i.i.d. sample from a Binomial(m, p) distribution, where m and p are known. Let $\hat{M}$ denote the sample median.

(a) **Jackknife Replications:** Define the jackknife replications $\hat{M}_{(-i)}$ for $i = 1, \ldots, n$. Describe the procedure to obtain the jackknife estimate of the median, $\hat{M}_{jack}$.

(b) **Jackknife Bias:** Provide the jackknife estimate of the bias for the sample median. Discuss how the discrete support of the Binomial distribution might result in a bias estimate of exactly zero.

(c) **Jackknife Variance:** Provide the jackknife estimate of the variance for $\hat{M}$.

(d) **Consistency:** It is a well-known result that the Delete-1 Jackknife variance estimator is inconsistent for the sample median.

  (i) Explain why the non-smooth nature of the median causes this inconsistency.

  (ii) Contrast this with the jackknife's performance when estimating the variance of the sample mean.

Problem 3.

Let $X_1, X_2, \ldots, X_n$ be an i.i.d. sample from a $N(\mu, \sigma^2)$ distribution where σ^2 is known. We are interested in estimating the parameter $\theta = \max(0, \mu)$. For this problem, assume the true mean is $\mu = 0$, so the target parameter is $\theta = 0$. We use the plug-in estimator:

$$\hat{\theta} = \max(0, \bar{X})$$

1. **Theoretical Bias:** Show that the theoretical bias of $\hat{\theta}$ is $O(n^{-1/2})$. Specifically, prove that:
$$E[\hat{\theta}] = \frac{\sigma}{\sqrt{2\pi n}}$$
(Hint: Recall that if $\mu = 0$, the sample mean $\bar{X} \sim N(0, \sigma^2/n)$. Use the density of the normal distribution to integrate over the positive support.)

2. **Jackknife Bias Expression:** Calculate the jackknife estimate of bias.

3. **Asymptotic Jackknife Bias:** Let $E[\widehat{Bias}_{jack}] = (n-1)(E[\bar{\theta}_{(\cdot)}] - E[\hat{\theta}])$. Show that for large n, the expected Jackknife bias does not vanish but rather scales with $n^{-1/2}$. Prove that:
$$\lim_{n\to\infty} \sqrt{n}E[\widehat{Bias}_{jack}] = \frac{\sigma}{2\sqrt{2\pi}}$$
Hint: Use the large-n approximation $\sqrt{1 - \frac{1}{n}} \approx 1 - \frac{1}{2n}$.

4. **Failure of Convergence:** Using your results from parts (1) and (3), calculate the expected bias of the corrected estimator $\hat{\theta}_{BC}$ in the limit. Does the Jackknife successfully produce an unbiased estimator for θ as $n \to \infty$? Briefly explain why jackknife fails to decrease the bias in this case.

5. **The Bootstrap Distribution:** Let $\bar{X}^*$ be the mean of a bootstrap sample $X_1^*, \ldots, X_n^*$ drawn with replacement from the original data. The bootstrap estimator is $\hat{\theta}^* = \max(0, \bar{X}^*)$.
   (a) Suppose the observed sample mean is exactly $\bar{X} = 0$. Using the fact that $\bar{X}^*|\text{data} \approx N(\bar{X}, \sigma^2/n)$, describe the bootstrap distribution of $\hat{\theta}^*$.
   (b) Calculate the probability $P^*(\hat{\theta}^* = 0)$ in this case.

6. **Bootstrap Bias Correction:** The bootstrap estimate of bias is given by $B_{boot} = E^*[\hat{\theta}^*] - \hat{\theta}$.
   (a) For the case where $\bar{X} = 0$, calculate B_{boot} in terms of n and σ.
   (b) Compare B_{boot} to the results in Part 1 and Part 2. Why is the Bootstrap more successful at capturing the magnitude of the bias at the "corner" than the Jackknife?

Problem 4

Let $X_1, \ldots, X_n$ be i.i.d. $Uniform(0, \theta)$. We use the MLE $\hat{\theta} = X_{(n)}$.

1. **Theoretical Limit:** Show that the scaled error $n(\theta - \hat{\theta})$ converges in distribution to $Exponential(\theta)$ as $n \to \infty$.

2. **Bootstrap Inconsistency:** Let $\hat{\theta}^*$ be the maximum of a bootstrap sample.
   (a) Prove that $P^*(\hat{\theta}^* = \hat{\theta}) \to 1 - e^{-1} \approx 0.632$ as $n \to \infty$.
   (b) Explain why the presence of this discrete point mass at zero prevents the bootstrap distribution of $n(\hat{\theta} - \hat{\theta}^*)$ from being a consistent estimator of the true error distribution.

3. **Jackknife Comparison:** The Jackknife replications are $\hat{\theta}_{(-i)} = \max_{j \neq i} X_j$.

   (a) Calculate the jackknife bias corrected estimator $\hat{\theta}_{jack}^{bc}$.

   (b) Show that $n(\hat{\theta}_{jack}^{bc} - \hat{\theta})$ has the same distribution as $n(\theta - \hat{\theta})$. In other words jackknife captures the spacing between the true value and the estimate correctly.

Problem 5.

Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, 1)$. We are interested in estimating the parameter $\theta = I(\mu > 0)$, which takes the value 1 if the mean is positive and 0 otherwise. We use the plug-in estimator $\hat{\theta} = I(\bar{X} > 0)$.

(a) **The Plug-in Estimator**
Assume the true mean is $\mu = 0$.

   (i) Show that the distribution of $\hat{\theta}$ is Bernoulli(0.5) for all n.

   (ii) Calculate the true variance of $\hat{\theta}$. Does this variance converge to 0 as $n \to \infty$? Discuss what this implies about the consistency of $\hat{\theta}$ at the point of discontinuity.

(b) **The Jackknife: False Certainty**
Let $\hat{\theta}_{(-i)} = I(\bar{X}_{(-i)} > 0)$ be the i-th jackknife replication, where $\bar{X}_{(-i)}$ is the sample mean excluding X_i.

   (i) Show that $\bar{X}_{(-i)} = \bar{X} + \frac{\bar{X} - X_i}{n-1}$.

   (ii) For a fixed $\bar{X} \neq 0$, show that as $n \to \infty$, the probability that $\hat{\theta}_{(-i)} = \hat{\theta}$ for all $i = 1, \ldots, n$ converges to 1.

   (iii) Based on this, what is the resulting Jackknife variance estimate $\widehat{\text{Var}}_{\text{jack}}(\hat{\theta})$ for large n? Explain why the Jackknife fails to capture the actual instability of the estimator.

(c) **The Bootstrap: Persistent Uncertainty**
The bootstrap estimate of the parameter is defined as the expectation of the indicator function under the bootstrap distribution: $\hat{\theta}_{\text{boot}} = E^*[I(\bar{X}^* > 0)]$.

   (i) Using the normal approximation $\bar{X}^* \sim N(\bar{X}, 1/n)$, show that the bootstrap estimate is:

$$\hat{\theta}_{\text{boot}} = \Phi(\sqrt{n}\bar{X})$$

   where Φ is the standard normal CDF.

   (ii) Calculate the limiting distribution. Does the variance of the bootstrap estimate go to zero as n increases?

   (iii) Contrast the Jackknife and Bootstrap results. Why is the "smoothing" property of the Bootstrap both a benefit (in providing a non-zero variance) and a limitation (in failing to converge to the truth) in this non-smooth case?